



BITS Pilani
Pilani Campus

Module 2: Mechanical Separations (Chapter 29)

Lecture 1

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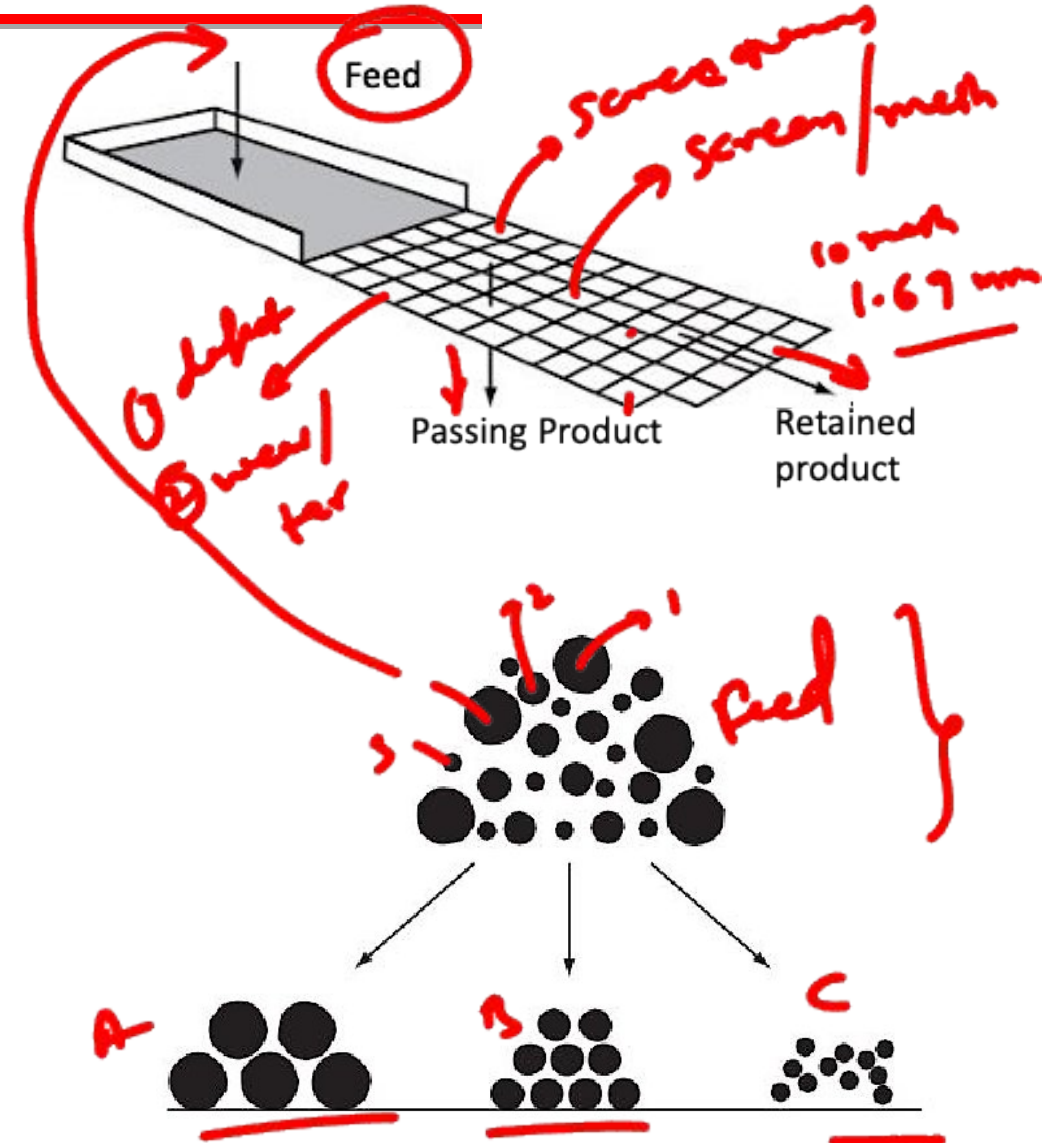
Lecture: Screening and Screen effectiveness

- Mechanical separations are applicable to heterogeneous mixtures, not to homogeneous solutions.
- The techniques are based on physical differences between the particles such as size, shape, or density.
- They are applicable to separating solids from gases, liquid drops from gases, solids from solids, and solids from liquids.

- General methods of mechanical separation:
 - the use of a sieve (**Screening**), septum, or porous membrane (**Filtration**) such as a screen or a filter which retains one component and allows the other to pass
 - the utilization of differences in the rate of sedimentation of particles or drops as they move through a liquid or gas.
 - Utilizing surface, electrical, or magnetic properties of particles.

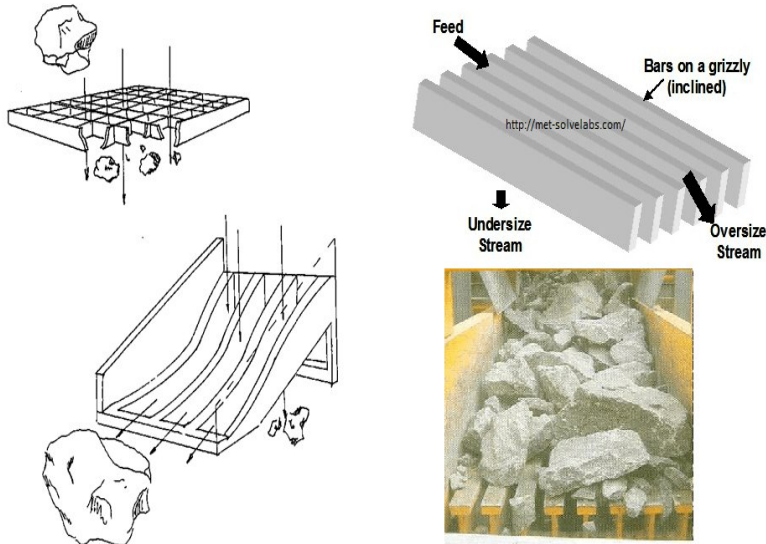
Screening

- Screening is a method of separating particles according to size alone.
- In industrial screening the solids are dropped on, or thrown against, a screening surface.
- The undersize or fines pass through the screen openings; oversize or tails do not.
- Applications of screening:
 - Mining and mineral processing, food, chemical, agriculture, plastics, recycling, and pharmaceuticals.

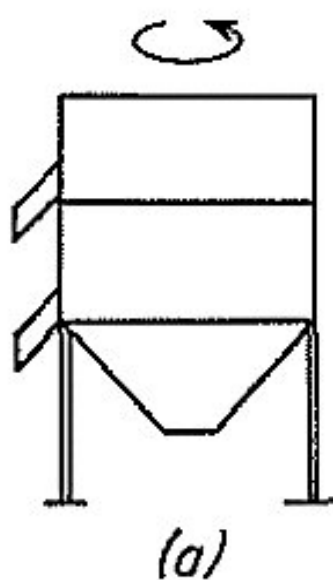


- Screening is done with the help of screening equipment
- Particles drop through openings with the help of gravity
- In few equipment, they are pushed through by a brush or centrifugal force
- Coarse particles can easily drop through large openings in stationary surface
- Fine particles needs to be agitated to pass through
- There are two methods of agitation
 - Shaking
 - Gyrating
- Typical motion of different screening equipment discussed in next slide

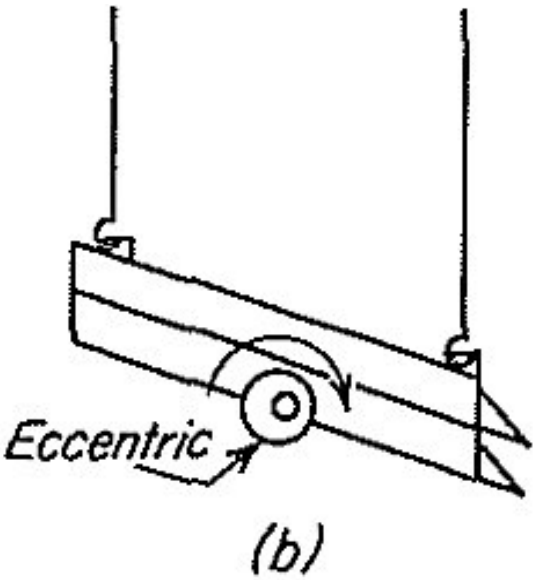
Motions of Screens



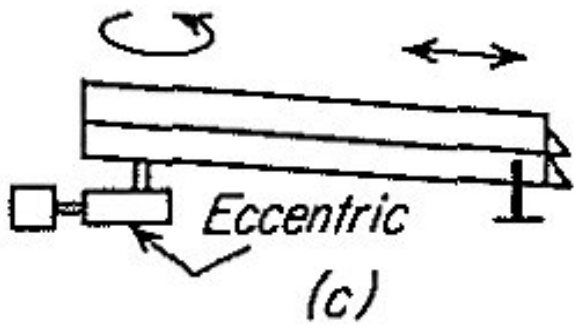
Stationary inclined screens and grizzlies



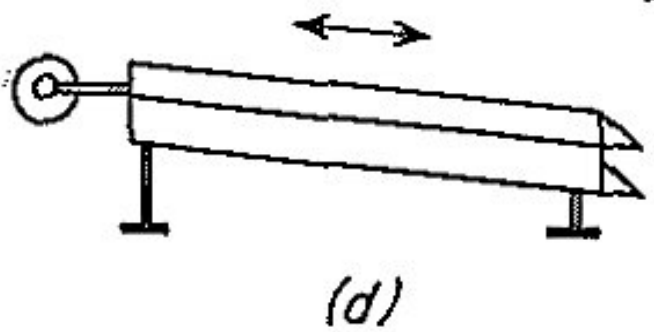
Gyrations in horizontal plane



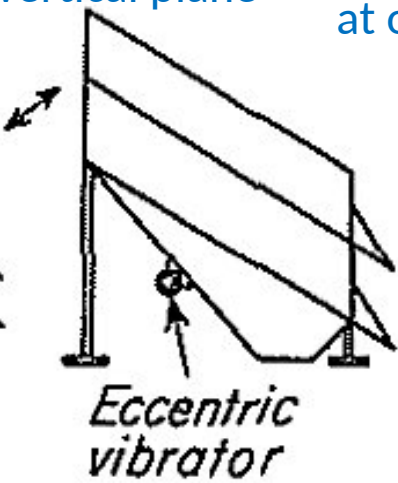
Gyrations in vertical plane



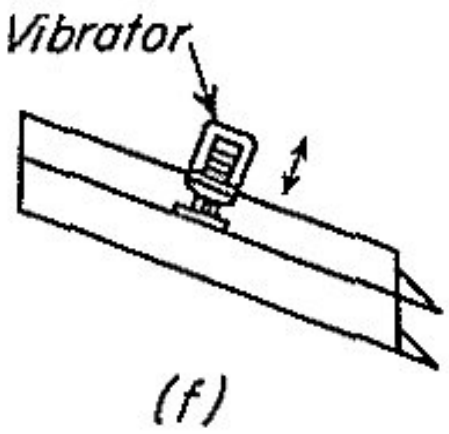
Gyrations at one end, shaking at other



Shaking



Mechanically vibrated

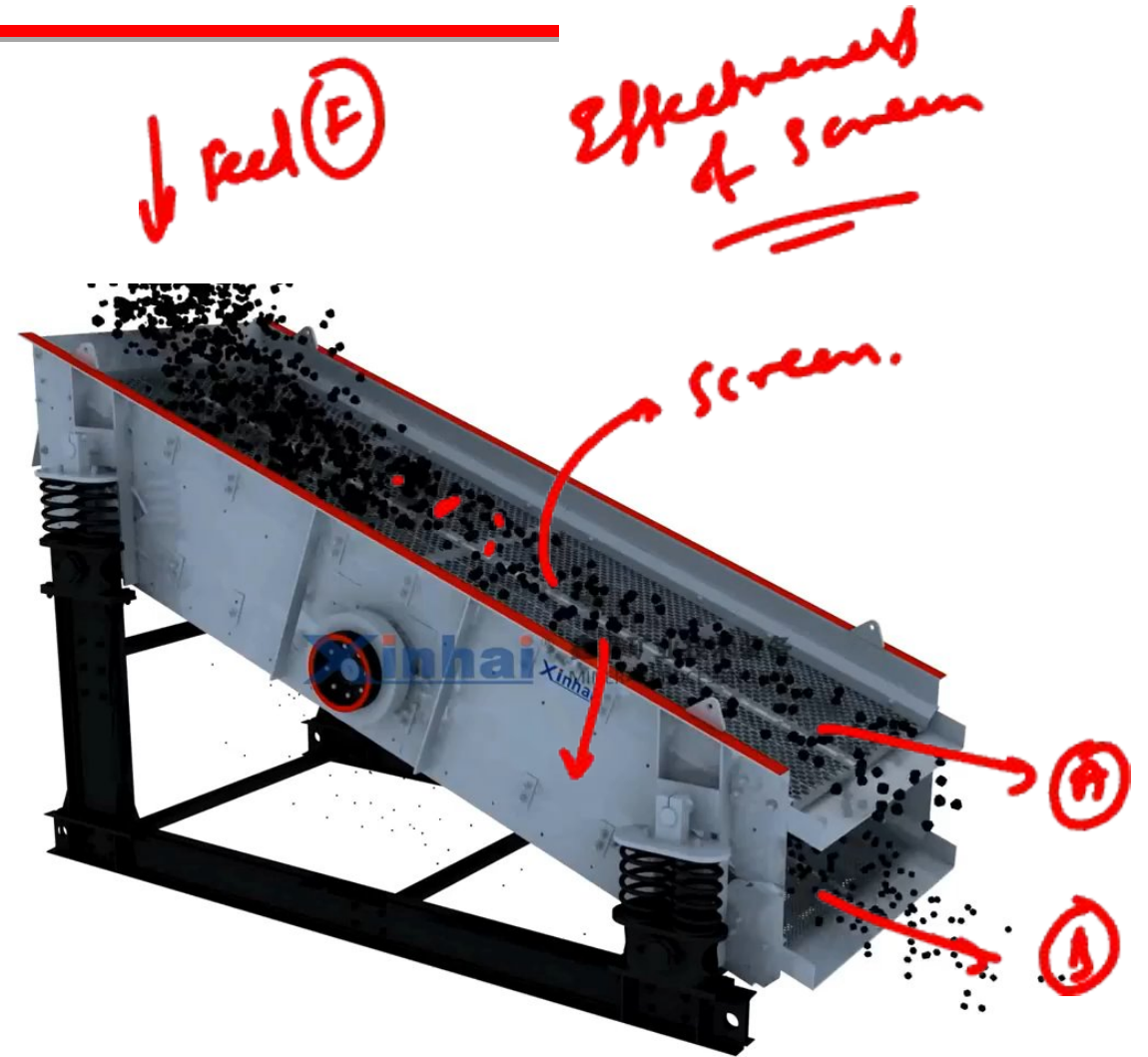


Electrically vibrated

Screening Equipment

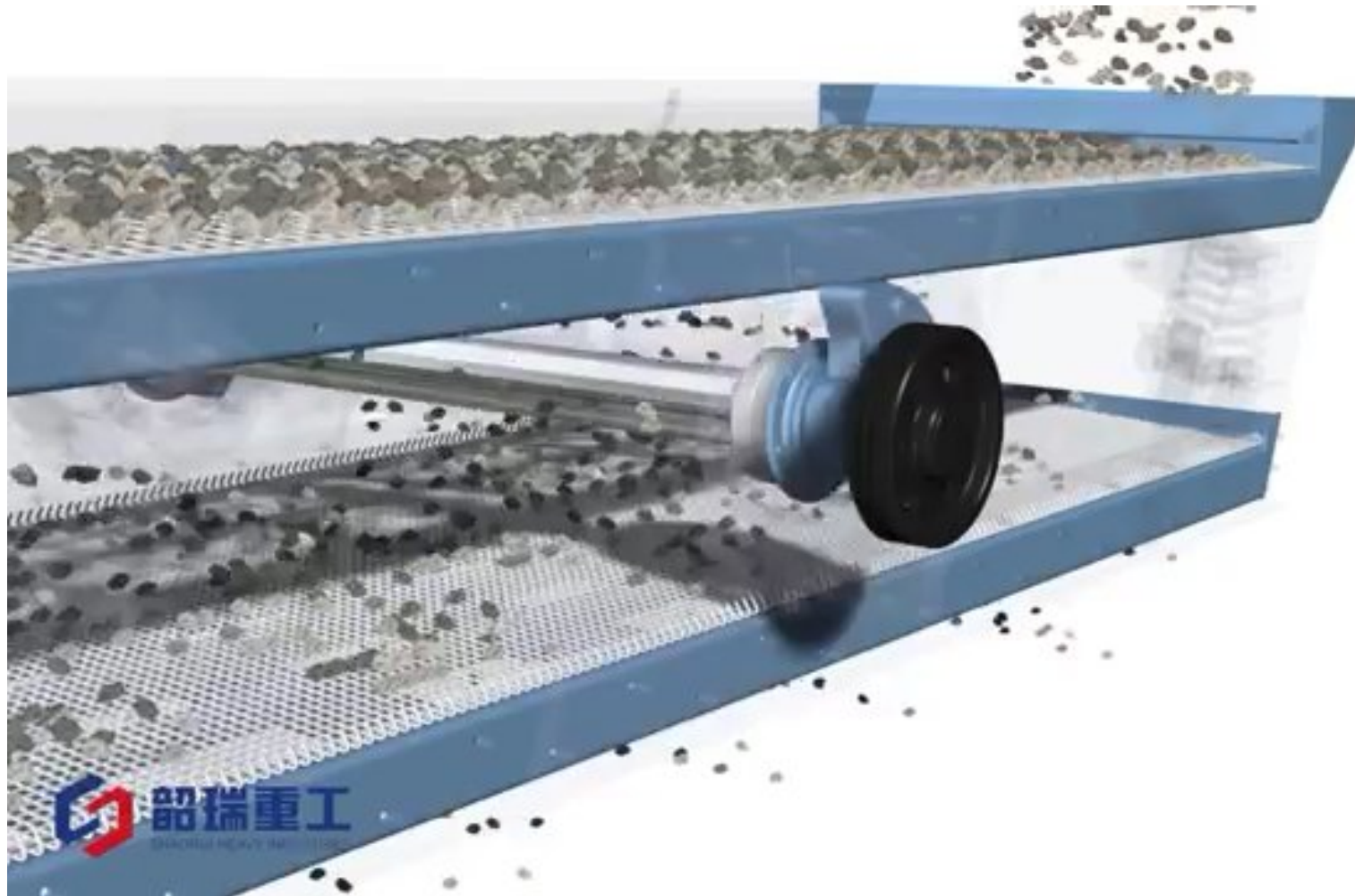


- Stationary screens and grizzlies
 - Coarse feed from primary crusher
- Gyrating screens
- Vibrating screens



<https://www.youtube.com/watch?v=JOk8oyWnrsg>

Screening Equipment



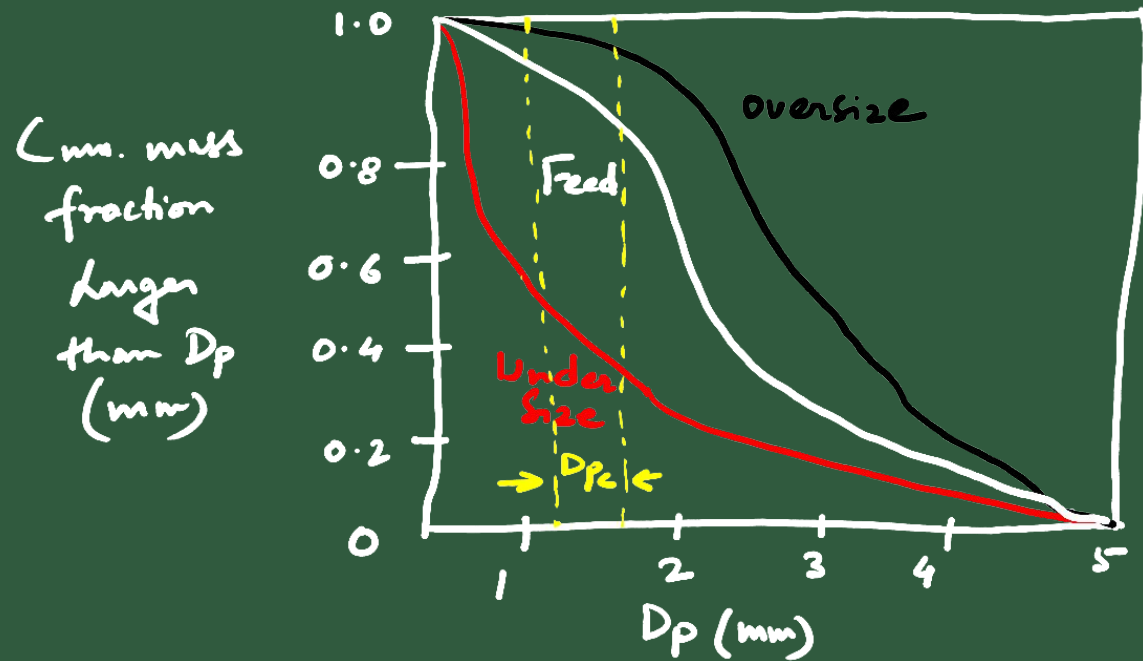
<https://www.youtube.com/watch?v=WoKRYKNvLpk>

Comparison of ideal and actual screens



- Objective of a screen
 - To accept a feed containing a mixture of particles of various sizes and separate into two fractions: oversize and undersize
 - Both may be products
- Ideal screen:
 - would sharply separate the feed mixture in such a way that the smallest particle in the overflow would be just larger than the largest particle in the underflow.
- Cut diameter D_{pc} :
 - marks the point of separation between the fractions
 - Chosen close to be equal to the mesh opening of the screen

Actual Screen $\rightarrow$ do not give perfect separation about cut diameter



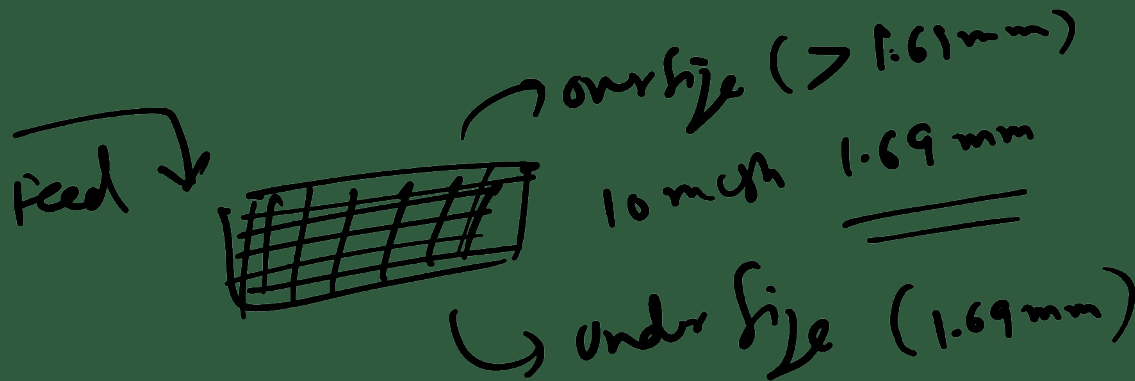
$D_{pc} \Rightarrow$ Cut point Diameter

point of separation b/w two size fraction

$D_{pc} \Rightarrow$ Mass opening of screen.

Commercial Screen
Poorer Separation

Testing Screen
Good Separation

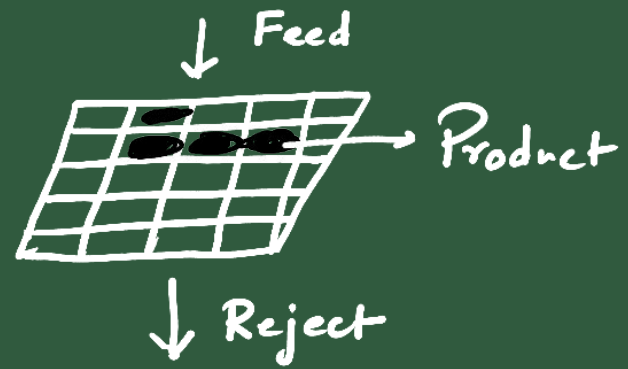


Material Balance over Screen

$F \rightarrow$ Mass flow rate of feed

D → Overflow Mass flow rate

B $\rightarrow$ Underflow Mass flow rate



Overall Mat. Balance

$$F = D + B \quad - (1)$$

x_F, x_D, x_B → Mass fraction of oversize Particles in Feed, overflow & underflow

$[1-x_f, 1-x_o, 1-x_B]$ " " Underize "

Overhze Mat. Balance

$$F x_F = D x_D + B x_B \quad \text{--- (2)}$$

Eliminating B,

$$Fz_F = Dz_D + (F-D)z_B$$

$$F(x_f - x_B) = D(x_D - x_B) \Rightarrow$$

$$\frac{D}{F} = \frac{x_{AF} - x_B}{x_B - x_D} = \frac{\text{Overflow Mass flow rate}}{\text{Feed Mass flow rate}} \quad \dots (3)$$

If D is eliminated

$$F x_F = (F-B) x_D + B x_B$$

$$F(x_F - x_D) = B(x_B - x_D)$$

$$\frac{B}{F} = \frac{x_B - x_D}{x_F - x_D} = \frac{\text{Underflow Mass flow rate}}{\text{Feed mass flow rate}} \quad \dots (4)$$

Screen Efficiency — Screen Effectiveness

Measures Success of a Screen in closely separating oversize (A) & Under size (B) particles.

if 100% efficiency, All D in overflow
" B " underflow

$$\text{Screen Effectiveness} = \frac{\text{oversize Mat. in overflow}}{\text{Amt of A in feed}} = \frac{D x_D}{F x_F} = E_A$$

$$= \frac{\text{Under size Mat. in Underflow}}{\text{Amt of B in feed}} = \frac{B(1-x_B)}{F(1-x_F)} = E_B$$

Combined Overall Effectiveness

$$E = E_A E_B = \frac{D x_D}{F x_F} \frac{B(1-x_B)}{F(1-x_F)} = \frac{D}{F} \frac{B}{F} \frac{x_D}{x_F} \frac{(1-x_B)}{(1-x_F)}$$

Replacing $\frac{D}{F}$ & $\frac{B}{F}$ from (3) & (4)

$$E = \left(\frac{x_F - x_B}{x_D - x_B} \right) \left(\frac{x_F - x_D}{x_B - x_D} \right) \frac{x_D}{x_F} \frac{(1-x_B)}{(1-x_F)} = \frac{x_D}{x_F} \frac{(x_F - x_B)(x_D - x_F)(1-x_B)}{(x_D - x_B)^2 (1-x_F)}$$

$$= \frac{(x_F - x_B)}{-(x_B - x_D)} = \frac{(\frac{x_F - x_D}{x_B - x_D} - x_B)x_D}{x_B - x_D(1-x_F)x_F} \frac{(x_F - x_B)(x_D - x_F)}{(x_D - x_B)^2}$$

Rearranging

$$\left[E = \frac{x_D}{x_F} \frac{(1-x_B)}{(1-x_F)} \frac{(x_F - x_B)(x_D - x_F)}{(x_B - x_D)^2} \right]$$

ABOVE EQN MAY ALSO BE WRITTEN AS:-

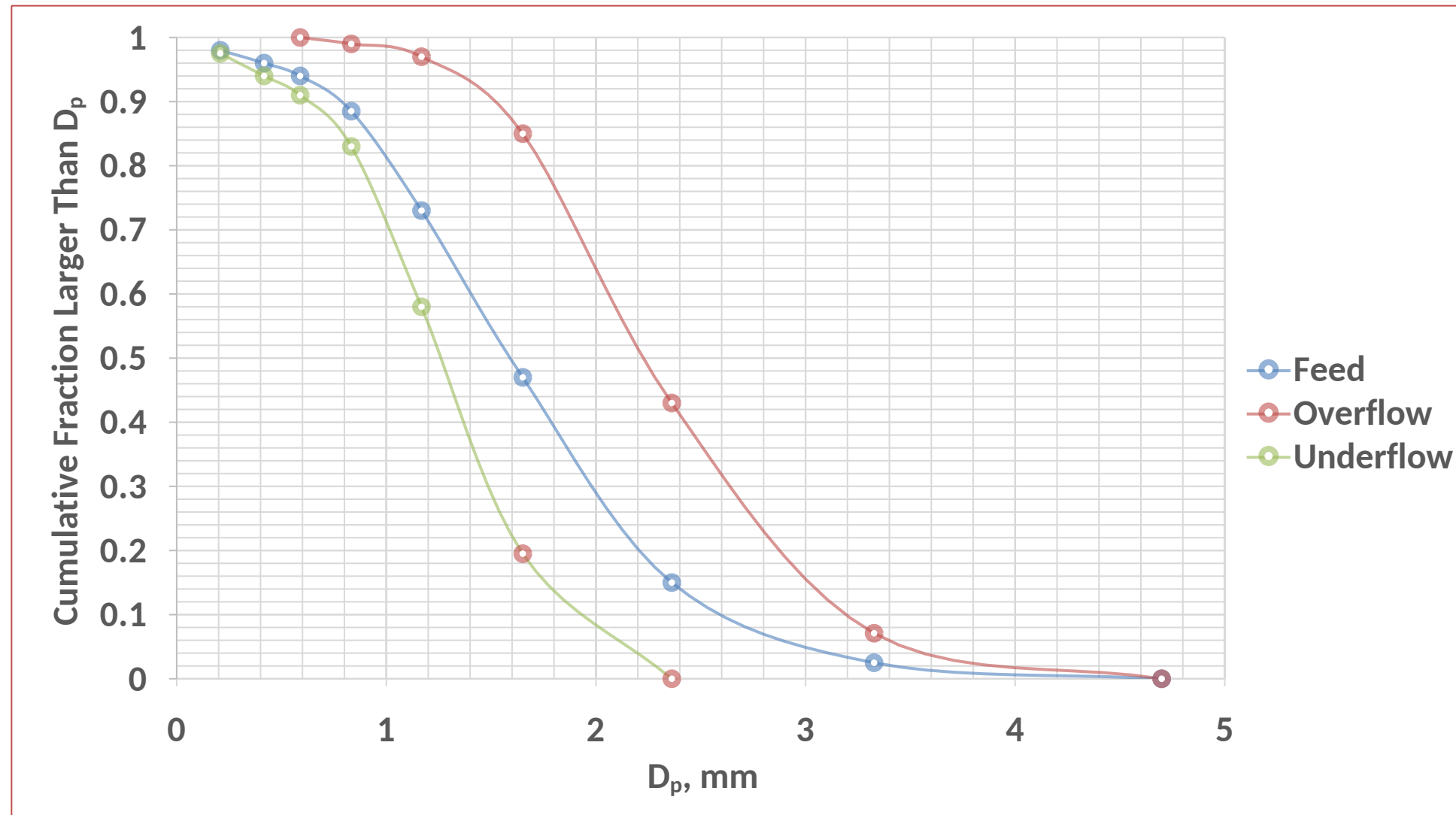
$$E = \frac{x_D}{x_F} \frac{x_F - x_B}{x_D - x_B} \left[1 - \frac{(1 - x_D)}{(1 - x_F)} \frac{(x_F - x_B)}{(x_D - x_B)} \right]$$

Screen analyses

- A quartz mixture having the screen analysis shown in Table is screened through a standard 10-mesh screen. The cumulative screen analysis of overflow and underflow are given in Table.
- Calculate
 - the mass ratios of the overflow and underflow to feed and
 - the overall effectiveness of the screen.

Mesh	D_p , mm	Cumulative fraction LARGER than D_p		
		Feed	Overflow	Underflow
4	4.699	0	0	
6	3.327	0.025	0.071	
8	2.362	0.15	0.43	0
10	1.651	0.47	0.85	0.195
14	1.168	0.73	0.97	0.58
20	0.833	0.885	0.99	0.83
28	0.589	0.94	1.00	0.91
35	0.417	0.96		0.94
65	0.208	0.98		0.975
Pan		1.00		1.00

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References



- Smith, J., Harriott, P., McCabe, W. (2005). Unit Operations of Chemical Engineering. Colombia: McGraw-Hill Education.
- Swain, A. K., Patra, H., & Roy, G. K. (2011). Mechanical operations. Tata McGraw Hill.